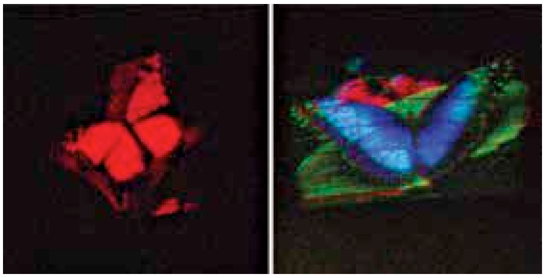
A new online video platform
YouTube and Vimeo have a new competitor – Wonder PL which provides ad-free exclusive lifestyle oriented video content. **http://dgit.in/WonderPL**

Android Wear
Google finally announced a special version of its Android OS tailored for wearable devices **http://dgit.in/WearDroid**



MIT researchers played out scene from Star Wars to test out their first hologram.

promise to revolutionise the world, we now look into the world of medicine and healthcare technology. Even as we invest billions in consumer technology for improving how well we enjoy life there is even more being invested in how well we live it and healthcare technology in human microbiome therapeutics is a huge part of that investment.

We've known for the longest time about the power of the gene but only recently have researchers of the current generation realised the practical benefit of larger biological entities like bacteria and fungi, popularly known as microbes. The human body itself is approximated to play host to over 10,000 microbial species and over 100 trillion microscopic organism which are present on the surface of the skin and also in our digestive systems - and they are a powerful part of human healthcare.

Scientists are catching up to understand the potential benefits microbes hold for humans with research such as the United States's government run National Institutes of Health's Human Microbiome Project. The four year old project has so far collected genomic data from nearly 200 microbes that could hold answers for human ailments. The project's goal is to catalogue and interpret the properties of 900 microbial organisms from human volunteers who are suffering from a variety of diseases. The investigation into how these microbes interact with human biology can hold critical answers towards creating new treatments and cures for a multitude of diseases, thus, saving millions of people.

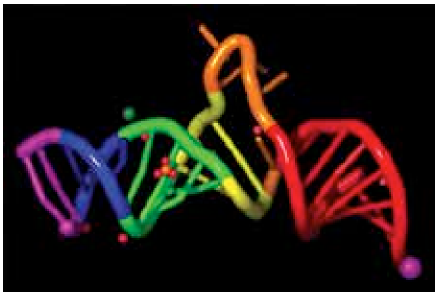
The private sector is equally invested in seeking out potential cures by researching microbes and companies like Second Genome have already begun their work analysing microbe DNA. By understanding the genetic mechanism of these microbes scientists at NIH and Second Genome believe that they can augment these trillions of organisms to enhance biological functions in humans like programmable organic nano-robots. By augmenting the microbes present in the digestive tract researchers can make human digestion more efficient, allowing for less wastage of nutrition, which could prove invaluable in poorer nations.

Other aspects of human biology like resistance to diseases, concerns of obesity and heart disease mitigation and cancer prevention can also be addressed if scientists are successful in controlling how microbes operate in the human body. And even though the field of human microbiome therapeutics is still in its infancy its potential to radically change human life is universally accepted.

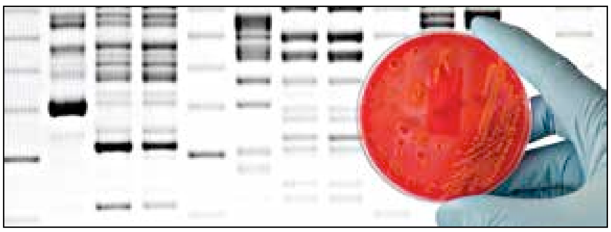
RNA-based Therapeutics
Technology that affects billions of people is usually invisible. Developments in fundamental sciences has become a natural part of existence like the harnessing of electricity and antibiotics. Without them life as we know it would be impossible. In this respect medical technology that creates life-saving drugs and treatments is one of the most important frontiers of innovation. And a new technique that is revolutionising how we provide treatment for millions of people is in the form of RNA-based therapeutics.

Ribonucleic acid or RNA is a fundamental part of any biological organism and instrumental in developing a new generation of medications that are more powerful and safer than currently available options. Even over-the-counter medications such as *Dispirin* work by effecting specialised enzymes (proteins) in the human body through artificial chemical compounds. By controlling the levels of these proteins in the human body we are able to change human biology such as relieving pain and reducing cholesterol. And the use of targeted RNA therapeutics is an extension of this process.

The next generation of medical research is concerned with narrowing the use of genetic research towards understanding more of how RNA-based treatments can benefit humanity. By delving deeper into the molecular dynamics of messenger RNA (mRNA) that give signals to enzyme creation in human cells, scientists have discovered a preemptive means of preventing diseases - microRNA (miRNA). According to current research microRNA control is the earliest step in the chain of biological interactions that can put an end to the very origin of deadly diseases. At present most medications only work towards inhibiting some microRNA but further developments in RNA-based therapeutics might change this forever by creating anti-microRNA drugs.



RNA based therapeutics



Genetic re-sequencing can allow scientists to repurpose everyday microbes for medical use.

Private firms like Regulus Therapeutics, Alnylam Pharmaceuticals and Isis Pharmaceuticals are driving the research in this field and are targeting a broad range of diseases such as hypercholesterolemia (pre-congenital high cholesterol), brain tumours, hepatitis and other life threatening diseases. One of the most recent creations by Isis targets microRNA that attack genetic hypercholesterolemia. Public research by the American government is also developing these techniques by experimenting with anti-miRNA drugs that target the replicating ability of the hepatitis virus. If they